Day 1 — The Moon and Why It Changes

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will explain, with a physical model, why the Moon appears to change shape over a ~29.5-day cycle, and will name the synodic month as the underlying period.

Materials

- A torch / flashlight (the brighter the better — bring spare batteries)
- A small white ball (a table-tennis ball or a styrofoam ball about 4 cm across)
- A darkened classroom (curtains drawn or lights off for ~10 min)
- Whiteboard
- Notebooks

Vocabulary introduced today

- *synodic month* — the ~29.53-day period from one new moon to the next; the basis of the lunar calendar.

(Sanskrit terms start tomorrow. Day 1 is observational and physical.)

Lesson flow

Warm-up (5 min)

- Opening question on board: *“Did you see the Moon last night? What shape was it?”*
- Take 4–5 students’ answers. Sketch each shape they describe on the board (don’t correct yet — just record).
- *“Why is it sometimes a full circle, sometimes a thin sliver, sometimes nothing at all? That’s what this module answers.”*

Core (20 min)

1. **The Moon is a ball lit by the Sun.** Hold up the white ball. *“This is the Moon — a rocky sphere, about a quarter the size of the Earth.”* Shine the torch on it from one side. *“The Sun lights up only one half of the Moon. Always. Half is day, half is night.”*

2. **Why the shape changes — the demo.** Have one student stand still in the centre of the room (= Earth). One student stands ~3 m away holding the torch on (= Sun). You walk slowly around the Earth-student in a wide circle, holding the ball at arm’s length so the Earth-student can see it.

   At each of 8 stopping points, ask: *“What fraction of the lit half of the ball can Earth see?”*

    - **Position 1 (Moon between Earth and Sun):** Earth sees the dark side. → *new moon*
    - **Position 2 (45° on):** A thin sliver visible on one side. → *waxing crescent*
    - **Position 3 (90°):** Half of the disc visible. → *first quarter*
    - **Position 4 (135°):** Most of the disc visible. → *waxing gibbous*
    - **Position 5 (Earth between Sun and Moon):** Earth sees the fully lit side. → *full moon*
    - **Positions 6–8:** Reverse — gibbous, half, crescent → *waning*
3. **Key insight on the board:**

   The Moon doesn't change. The angle from which we see it changes.
4. **The cycle takes ~29.5 days.** This is the **synodic month** — the time from one new moon to the next. Write the number 29.53 on the board.

5. **One question to plant:** *"Earth orbits the Sun in 365 days. The Moon orbits Earth in 29.5 days. Is the lunar 'month' the same as a calendar month?"* (No — and that's why we have leap years for the solar calendar, and a fancier system for the lunar one. We'll get to that on Day 5.)

Activity (15 min) — Pair sketching

In pairs:

1. One partner is "Sun" (holds an imaginary torch in one direction).
2. The other is "Earth" (stays still).
3. Take a small ball or a balled-up paper and walk it around Earth in 8 stopping positions.
4. At each position, the Earth-partner sketches what shape they see in their notebook.

After 7 minutes, swap roles.

By end of activity, each student has 8 small Moon-shape sketches in their notebook labelled new, waxing crescent, first quarter, waxing gibbous, full, waning gibbous, last quarter, waning crescent.

Wrap-up (5 min)

- **Exit ticket on a slip of paper, 2 min:** *"In one sentence: why does the Moon look different on different nights?"*
- Collect. Read 3 aloud quickly. Correct any "Earth's shadow makes the phases" answers gently — Earth's shadow only matters for *eclipses* (Day 7).
- **Tonight's homework:** Look at the Moon. Note the date, the time, the shape (sketch). Write down the cloud condition. This starts your **10-night Moon diary**.

Student-facing content

> **The Moon does not change. Our view of it changes.**
>
> The Sun lights up half of the Moon. As the Moon orbits Earth, we see different fractions of that lit half. Over ~29.5 days, the visible fraction grows from nothing (new

moon), to a half-disc (quarter), to a full disc (full moon), back to a half, and back to nothing.

This 29.5-day cycle is called the **synodic month** — the basis of the lunar calendar that we'll study in this module.

Homework

- **Tonight (and every night for 10 nights):** Look at the Moon. Sketch its shape in your workbook. Note the date, the time, and "clear / partly cloudy / overcast." If overcast, look it up on a Moon-phase app or NASA's published phase calendar.

Differentiation

- **One tier up:** Why is the synodic month (29.53 days) different from the *sidereal* month (27.32 days, the time for the Moon to return to the same star background)? Hint: Earth has moved around the Sun during the lunar month.
- **One tier down:** Skip the 8-position drill. Focus on 4: new, half (waxing), full, half (waning). Build the rest later in the week.

Teacher notes

- Some students will say "the Moon makes its own light." It does not — it reflects sunlight. Show them this physically by switching the torch off in the demo. The ball goes dark.
- Some will say "the Moon's far side is permanently dark." It is not. The Moon's far side gets just as much sunlight as the near side, on average — we just never see the far side from Earth because the Moon is tidally locked (the same face always faces us).
- For the diary, if students live in a heavily light-polluted city, the thin crescents may be hard to spot in twilight. They can use an app for verification — that's acceptable.

Citation(s) used in this lesson

- General reference: NCERT Science Class 6, Chapter 17 ("Stars and the Solar System"). No verse citation today.